# Inverse relations and functions

**1** $y = x$

**2**

**a** No **b** Yes **c** No **d** Yes

**e** No **f** Yes **g** No **h** Yes

**3**

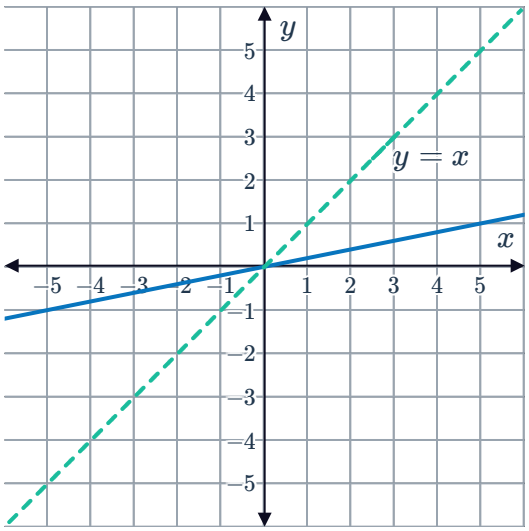
**a** Yes **b** No **c** Yes **d** Yes

**e** No **f** No **g** Yes **h** Yes

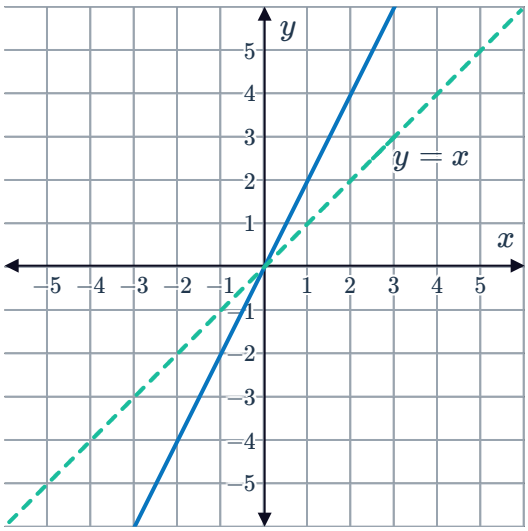
**i** Yes **j** Yes **k** No

**4**

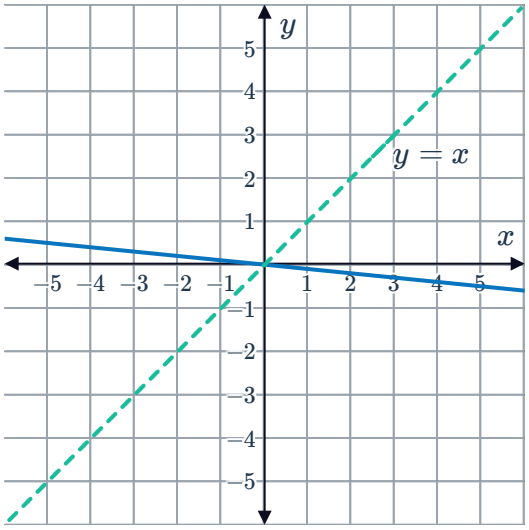
**a**



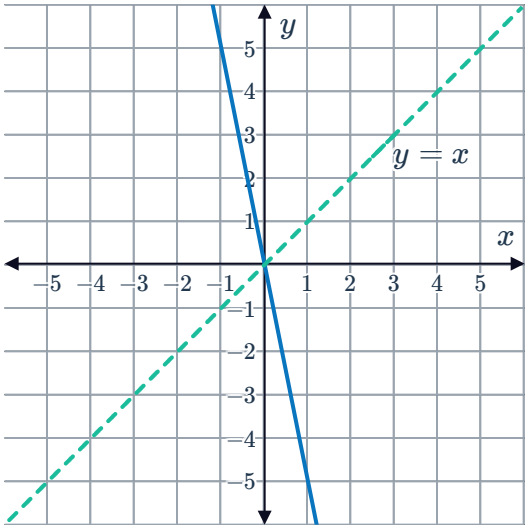

**b**




**c**

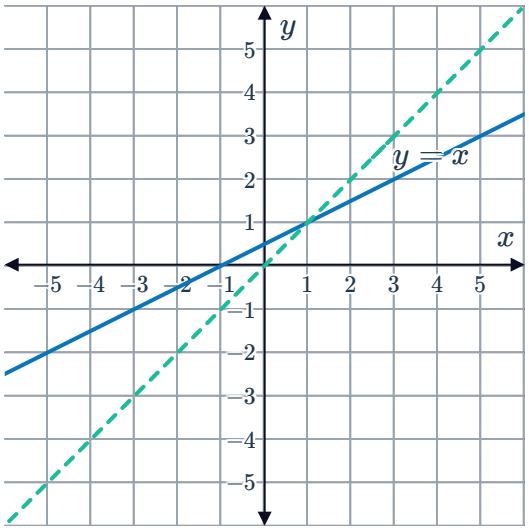



**d**

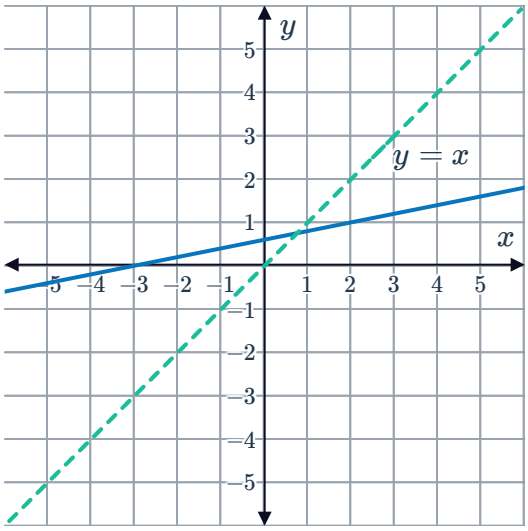



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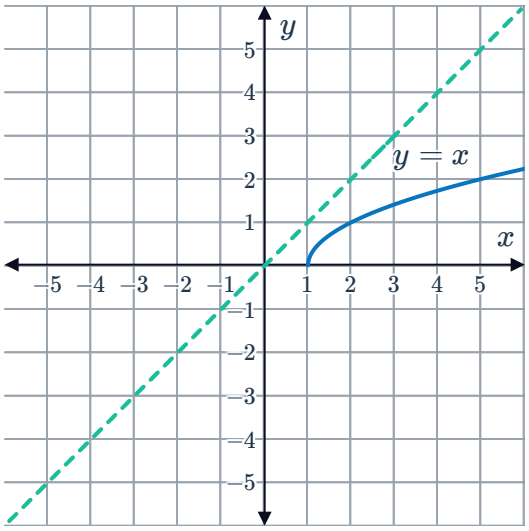
e



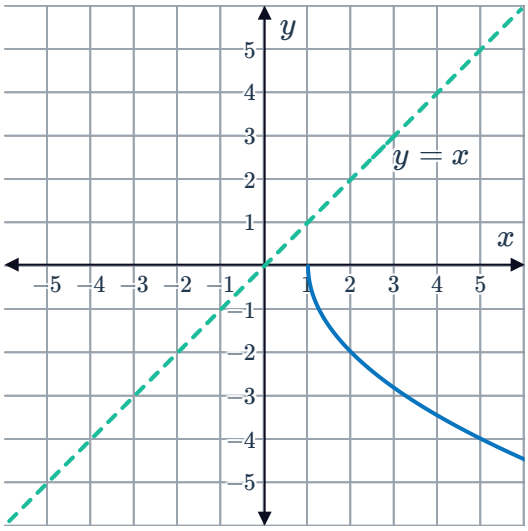

f



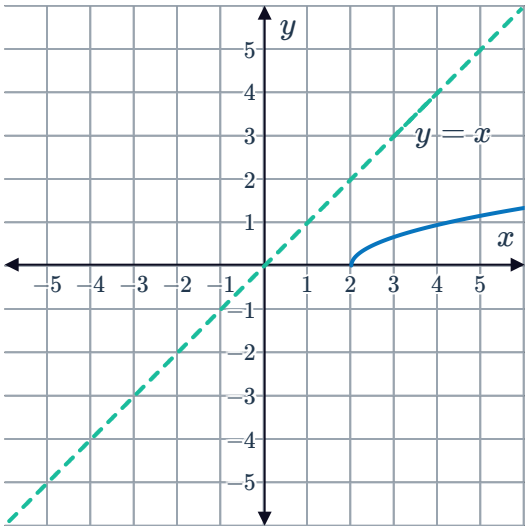

g



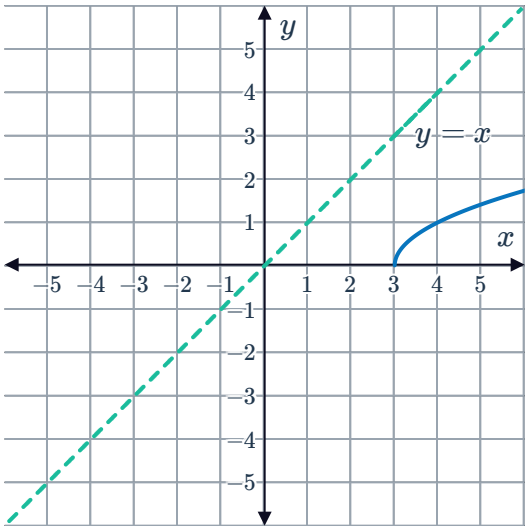

h




i




j




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**5**

**a** Function

**b** Relation

**6**

**a**

**i** $x$ **ii** $x$ **iii** Yes

**b**

**i** $x - 12$ **ii** $x - 4$ **iii** No

**c**

**i** $x$ **ii** $x$ **iii** Yes

**7**

**a** $y = \frac{x}{5}$

**b** $y = \frac{x + 8}{7}$

**c** $y = \frac{3}{4}(x + 5)$

**d** $y = \sqrt[3]{\frac{x}{6}}$

**e** $y = 8(1 - x)$

**f** $y = 6x + 7$

**g** $y = x^3 + 2$

**h** $y = \sqrt[3]{x - 3} + 4$

**i** $y = -\frac{2x}{x - 5}$

**j** $y = \frac{8}{x - 5} + 9$

**k** $y = \frac{8}{5x} - \frac{3}{5}$

**l** $y = \sqrt[3]{\frac{7}{x + 3}} + 8$

**m** $y = \frac{\sqrt[3]{\frac{x + 7}{4}} + 2}{9}$

## Domain and range

**8**

**a** $\{-3\}$

**b** $\{6\}$

**c** $(6, -3)$

**d** $\{6\}$

**e** $\{-3\}$

**f** The domain of $f$ is equal to the range of $f^{-1}$, and the range of $f$ is equal to the domain of $f^{-1}$.

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**9**

**a** $\{6, -5\}$ **b** $\{6, -5\}$

**10**

**a** $\{5, 7, 8\}$ **b** $\{9, -4, 2\}$

**c** $\{(9, 5), (-4, 7), (2, 8)\}$ **d** $\{9, -4, 2\}$

**e** $\{5, 7, 8\}$

**11**

**a** $a = -\frac{1}{2}, b = \frac{5}{2}$ **b** $[5, 0]$ **c** $[-5, 5]$

**12**

**a**

**i** $f^{-1}(x) = \frac{1}{5}(x - 15)$ **ii** $(-\infty, \infty)$

**iii** $(-\infty, \infty)$

**b**

**i** $f^{-1}(x) = \frac{1}{3}(4 - x)$ **ii** $(-\infty, \infty)$

**iii** $(-\infty, \infty)$

**c**

**i** $f^{-1}(x) = \frac{1}{8}(x + 9)$ **ii** $[-41, 7]$

**iii** $[-4, 2]$

**d**

**i** $f^{-1}(x) = \sqrt{x}$ **ii** $[0, \infty)$

**iii** $[0, \infty)$

**e**

**i** $f^{-1}(x) = \sqrt{16 - x^2}$ **ii** $[0, 4]$

**iii** $[0, 4]$

**13**

**a**

**i** $f(x) = \frac{2x}{3} - 2$ **ii** $g(x) = \frac{3x}{2} + 3$

**b**

**i** $x$ **ii** $x$

**c**

**i** True **ii** True **iii** False **iv** True

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**14**

**a** $[-4, 9]$ **b** $[-4, 5]$ **c** $[-4, 5]$ **d** $[-4, 9]$

**e** False

**15**

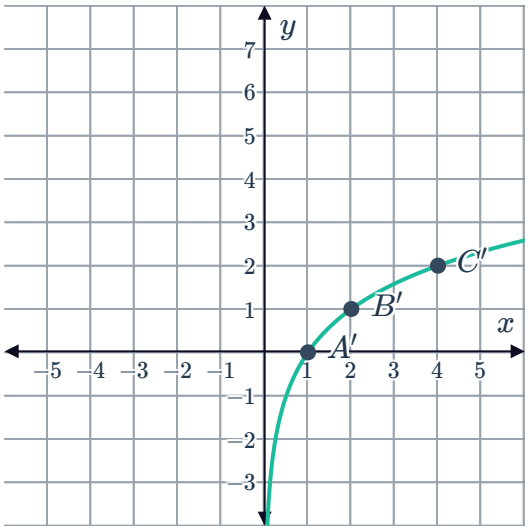
**a**

i $(0, 1)$ ii $(1, 2)$ iii $(2, 4)$

**b**

i $(1, 0)$ ii $(2, 1)$ iii $(4, 2)$

**c**




**16**

**a** $(-\infty, \infty)$ **b** $(0, \infty)$

**c** $(0, \infty)$ **d** $(-\infty, \infty)$

**e** The range of $f^{-1}$ **f** The domain of $f^{-1}$

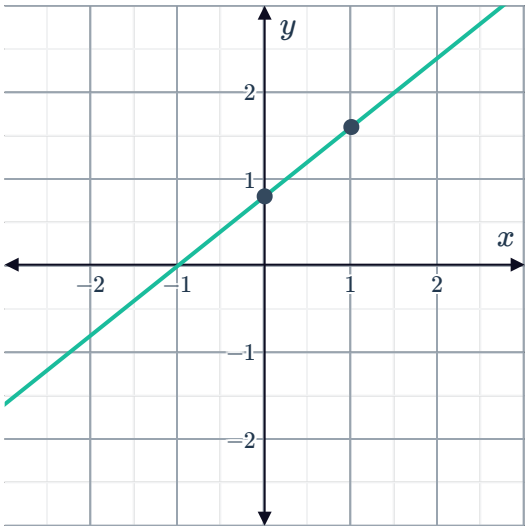
**17**

**a** $x \geq 2$ **b** $y \geq -5$ **c** $y = x^2 - 4x + 9$

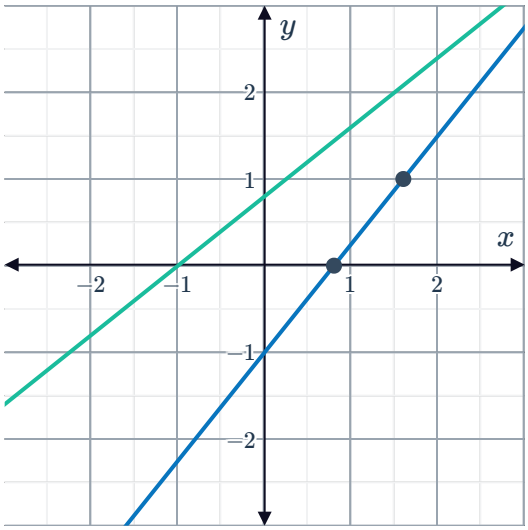
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**18**

**a**




**b**




**c** $(-\infty, \infty)$

**d** $(-\infty, \infty)$

**e** $(-\infty, \infty)$

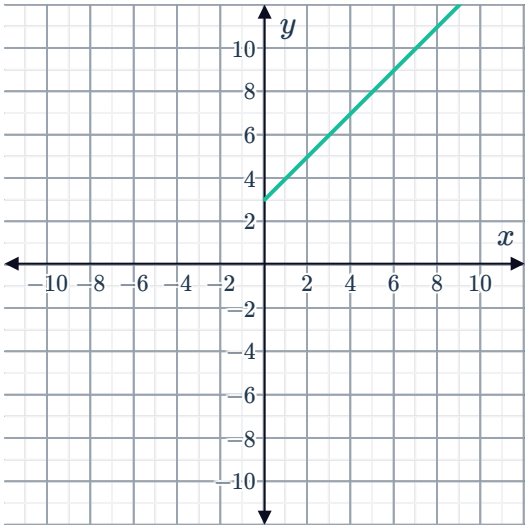
**f** $(-\infty, \infty)$

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**19**

**a**

**i**

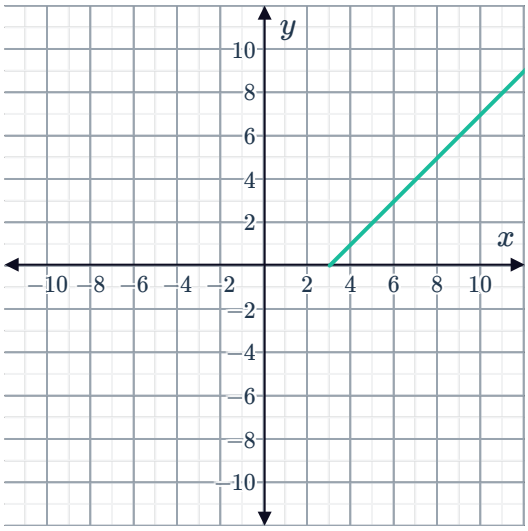



**ii** $f^{-1}(x) = x - 3$

**iii** $[3, \infty)$

**iv** $[0, \infty)$

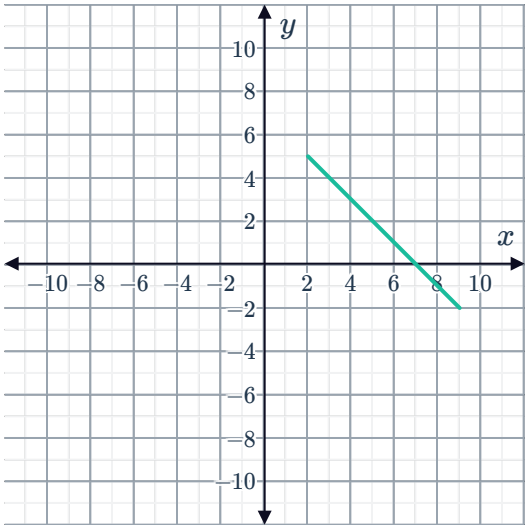
**v**




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**b**

**i**

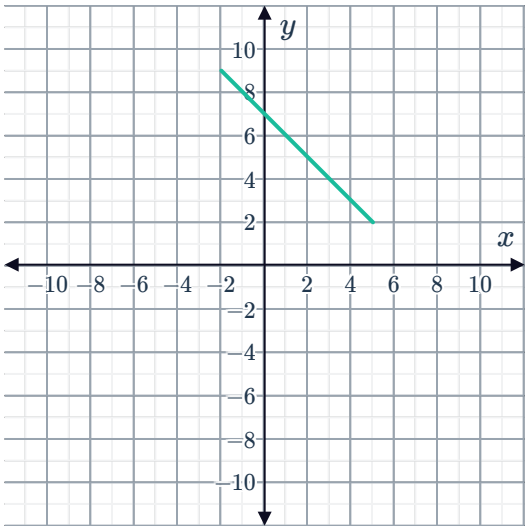



**ii** $f^{-1}(x) = 7 - x$

**iii** $[-2, 5]$

**iv** $[2, 9]$

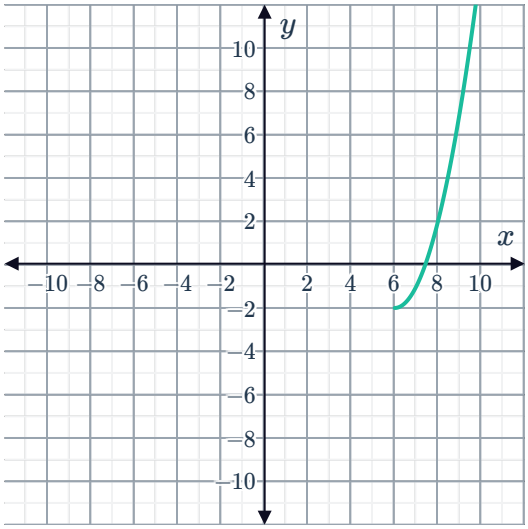
**v**




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**c**

**i**

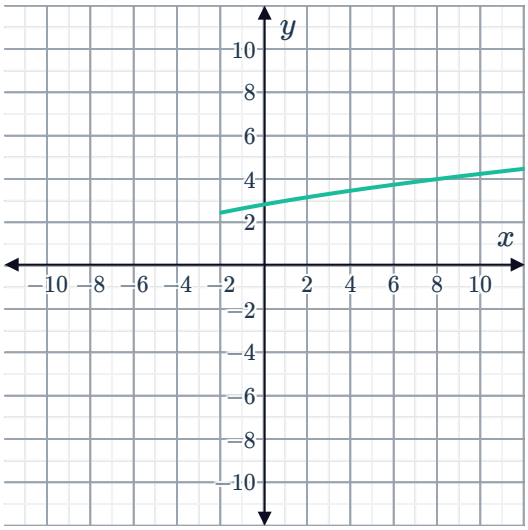



**ii** $f^{-1}(x) = \sqrt{x+2} + 6$

**iii** $[-2, \infty)$

**iv** $[6, \infty)$

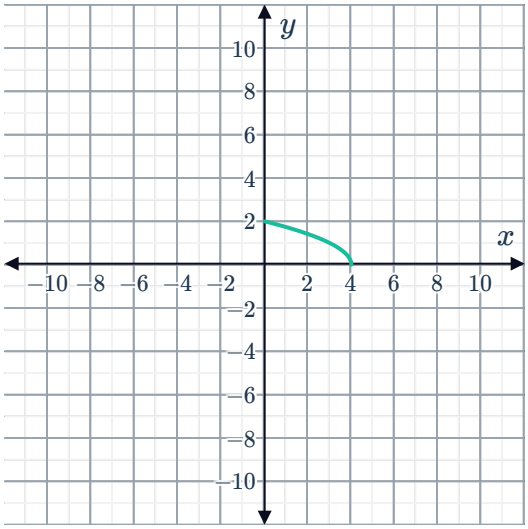
**v**




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**d**

**i**

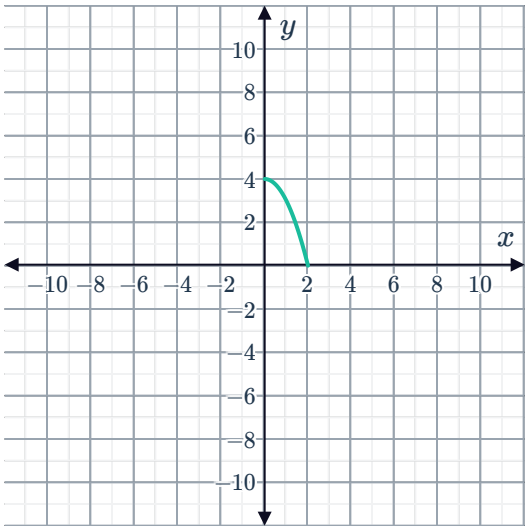



**ii** $f^{-1}(x) = 4 - x^2$

**iii** $(0, 2]$

**iv** $[0, 4)$

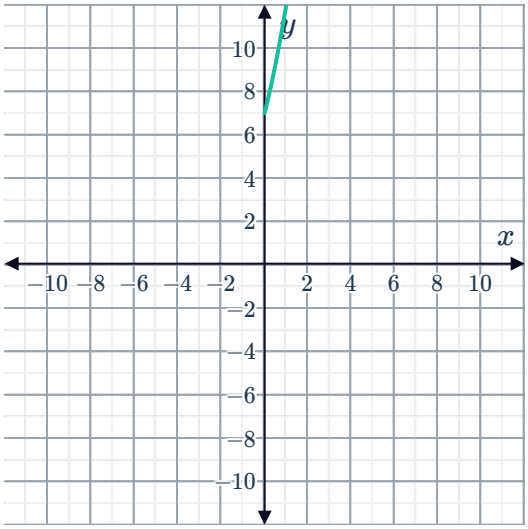
**v**




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**e**

**i**

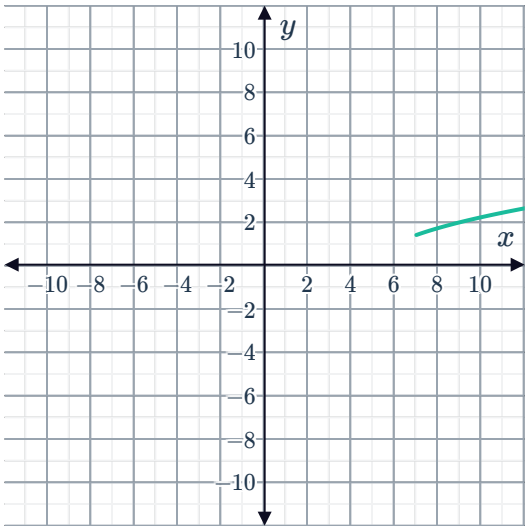



**ii** $f^{-1}(x) = \sqrt{x-3} - 2$

**iii** $[7, \infty)$

**iv** $[0, \infty)$

**v**




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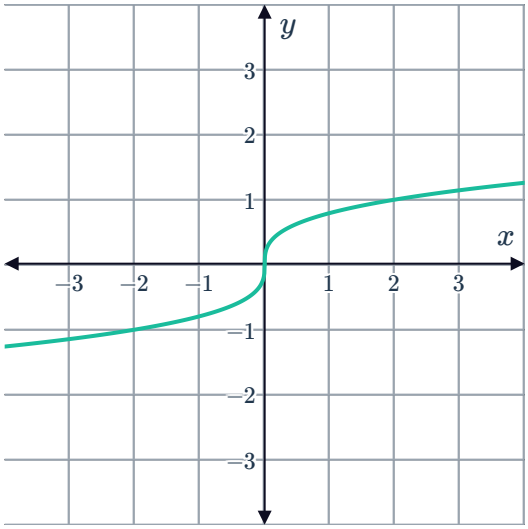
**20**

**a** Yes

**b** All real $x$

**c** All real $y$

**d**




**21**

**a** No

**b** All real $x$

**c** $x > 2$ and $x < -2$

**d** $y > 2$ or $y < -2$

**22**

**a** $T = 1.07A$

**b** $A = \dfrac{T}{1.07}$

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